

POLITECNICO
MILANO 1863

DEPARTMENT OF MECHANICAL
ENGINEERING

ESE05

Additional exercises

Name Surname



DIPARTIMENTO DI ECCELLENZA
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Milano

Exercise 1

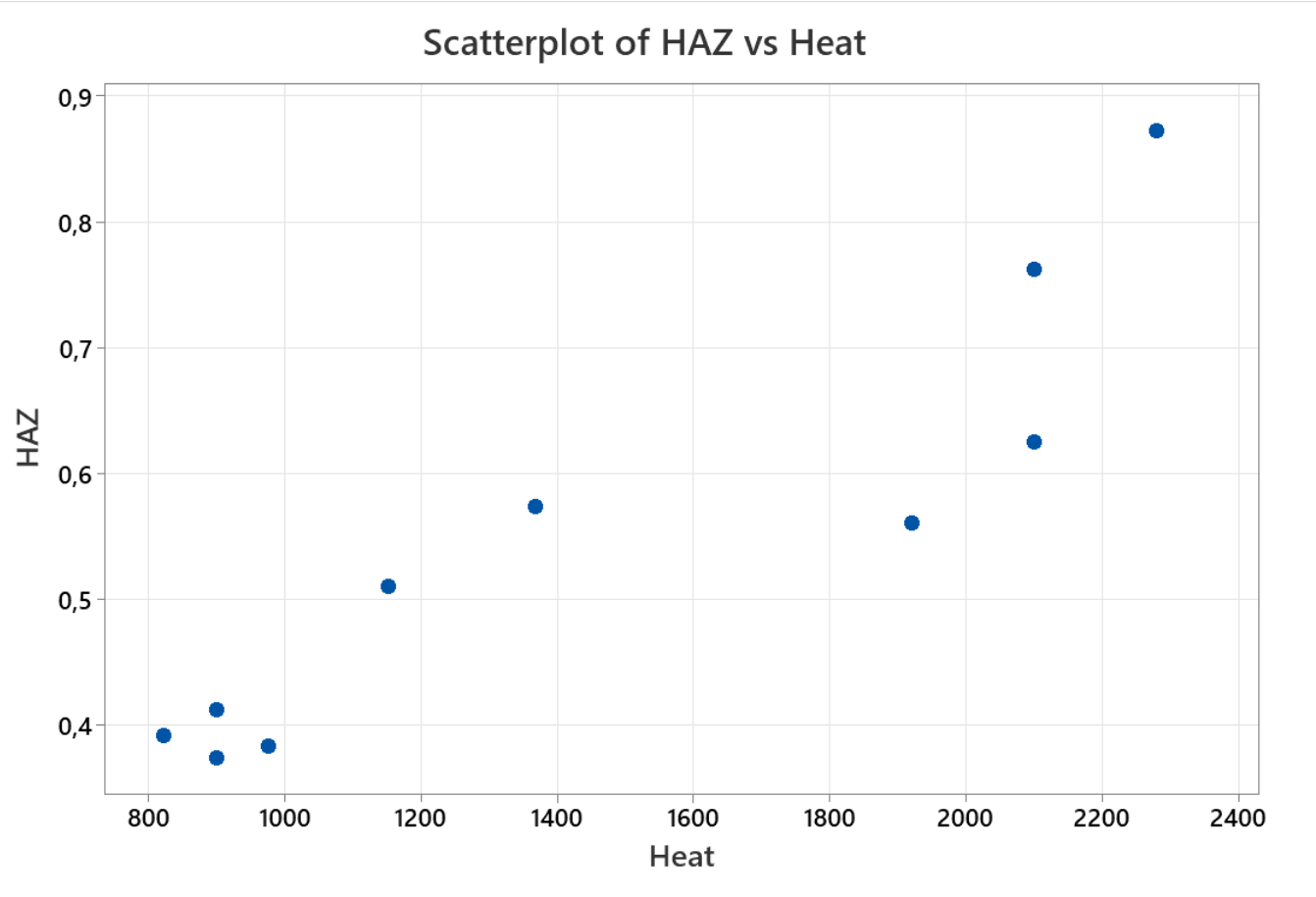
We are investigating the effect of the influence of the heat input (J/cm²) of a CW 1.5kW CO₂ laser on the heat affected zone width (W_{HAZ}) [mm] of a butt-welding of a medium carbon steel. The results are in the “HAZ.csv”

1. Suggest a model for the data.
2. Compute the prediction and confidence Heat input equal to 1260 J/cm²

Heat Input[J/cm ²]	W_{HAZ} [mm]
1920	0,561
2280	0,872
823	0,392
977	0,384
1152	0,453
1368	0,569
1152	0,511
1368	0,574
2100	0,625
900	0,375
2100	0,762
900	0,413

1.1 Suggest a model for the data.

Graph → Scatterplot = HAZ vs Heat



By looking at the scatterplot we can see a clear linear trend between the width (y) and the regressor (HAZ).

No information on time order, so we cannot check autocorrelation.

We fit a linear model to the data.

1.1 Suggest a model for the data.

Stat → Regression → Fit regression model → Response = HAZ, Continuous predictor = Heat

Storage: Fits and residuals

Graphs: Four in one

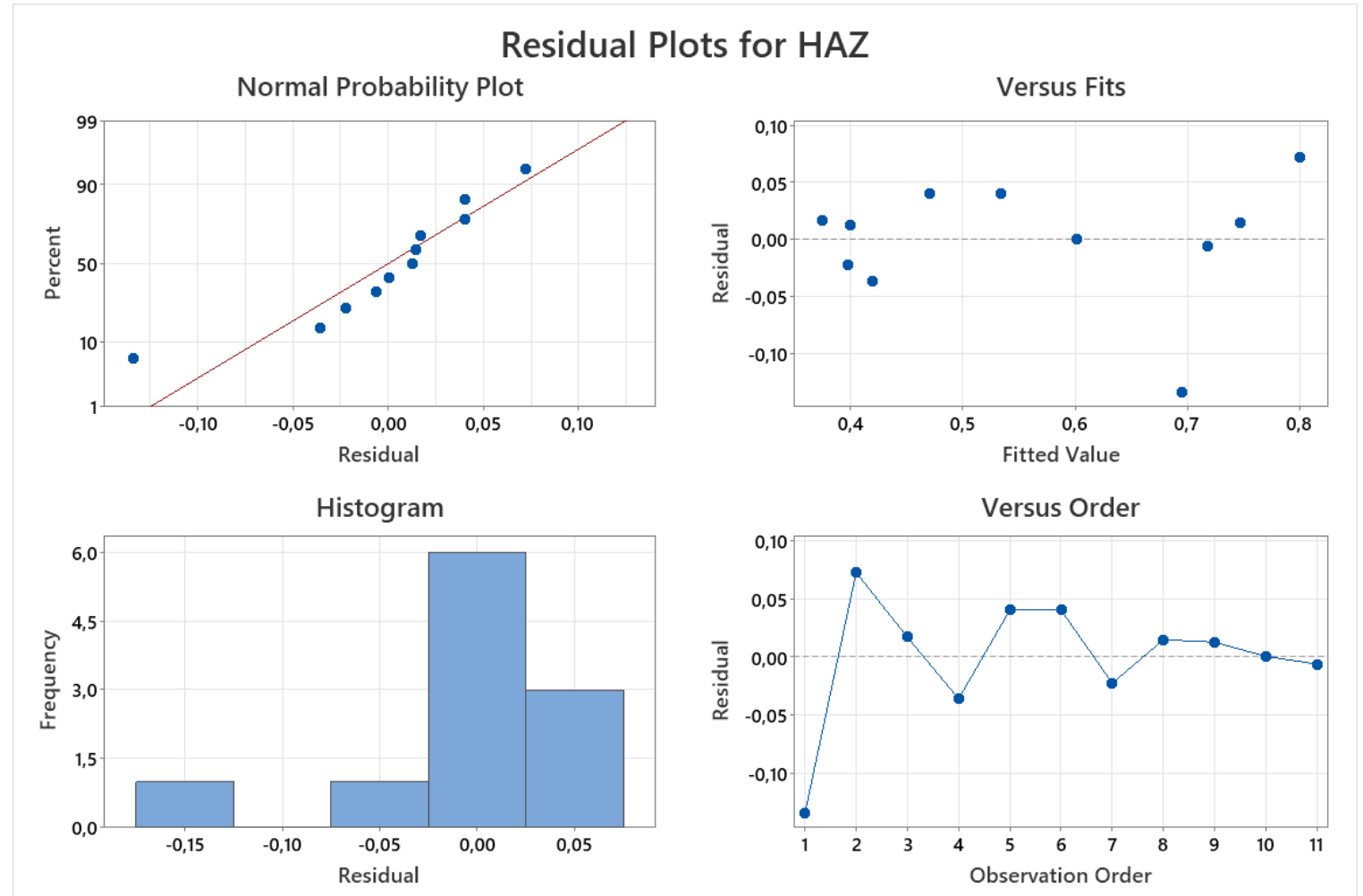
Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	0,25181	0,251813	78,60	0,000
Heat	1	0,25181	0,251813	78,60	0,000
Error	9	0,02884	0,003204		
Total	10	0,28065			

Model Summary

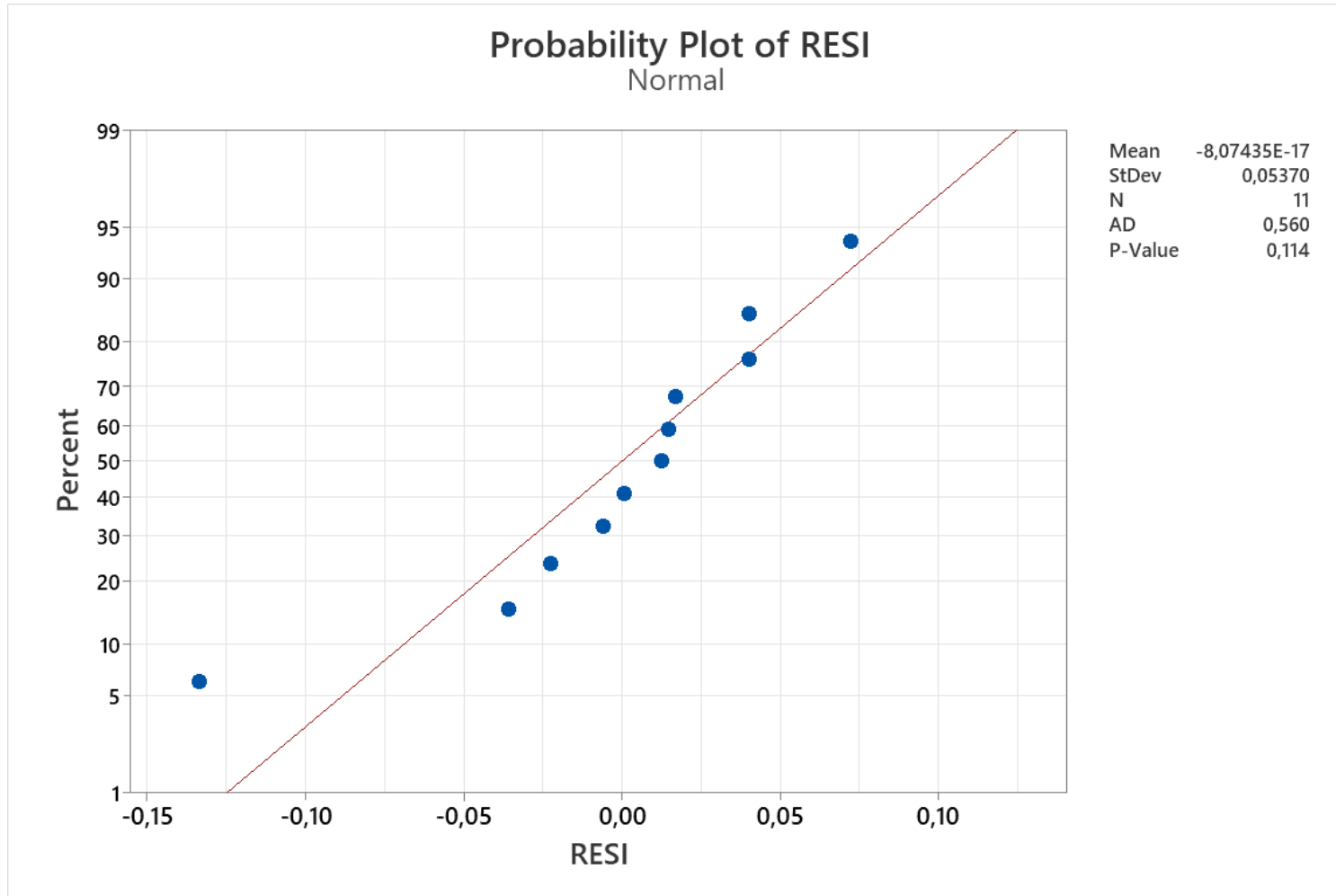
S	R-sq	R-sq(adj)	R-sq(pred)
0,0566030	89,73%	88,58%	84,08%

Residuals check



1.1 Suggest a model for the data.

Stat → Basic statistics → Normality test



The pvalue of the normality test is 0,114 ($> 0,05$) so we cannot reject the null hypothesis of normal assumption.

1.2 Compute the prediction and confidence Heat input equal to 1260 J/cm2

Stat → Regression → Predict.
Heat = 1260

Regression Equation

HAZ = 0,1350 + 0,000292 Heat

Settings

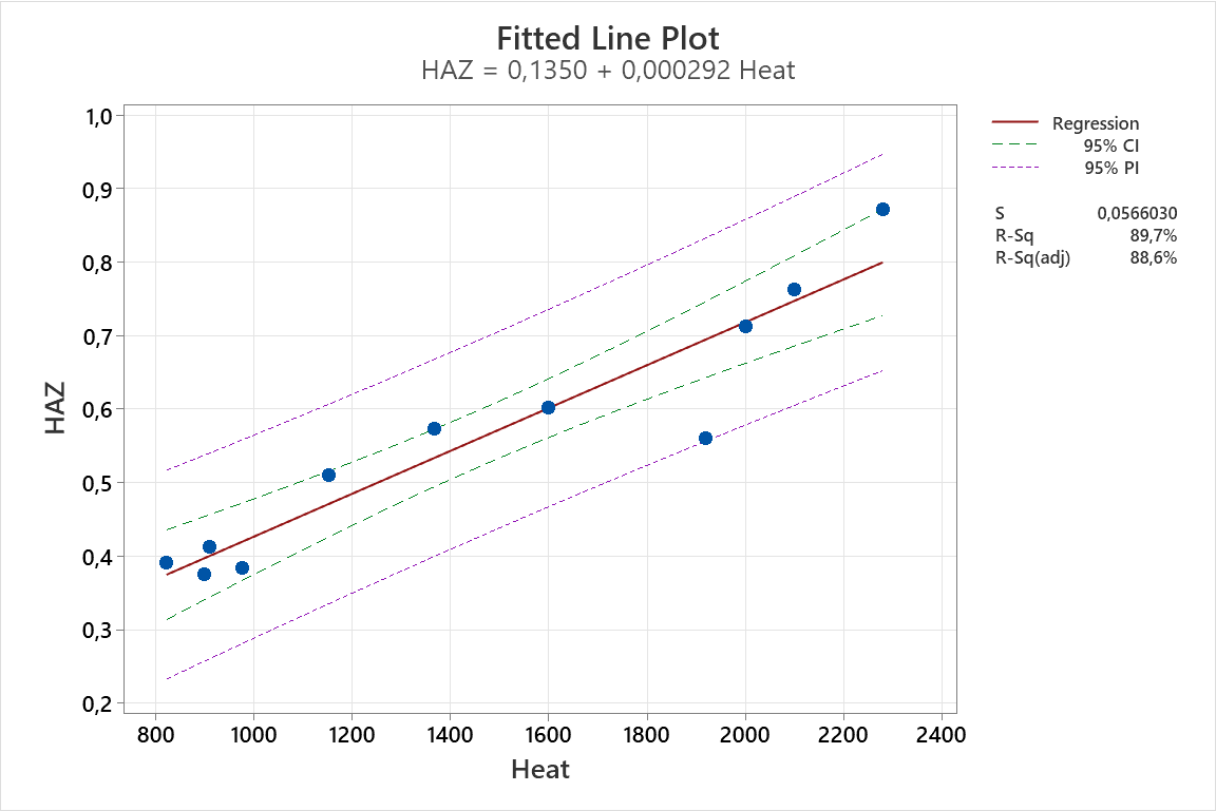
Variable Setting

Heat 1260

Prediction

Fit	SE Fit	95% CI	95% PI
0,502311	0,0182576	(0,461009; 0,543612)	(0,367770; 0,636852)

Stat → Regression → Fitted Line Plot
Y = HAZ, X = Heat
Options: Display Confidence and prediction intervals



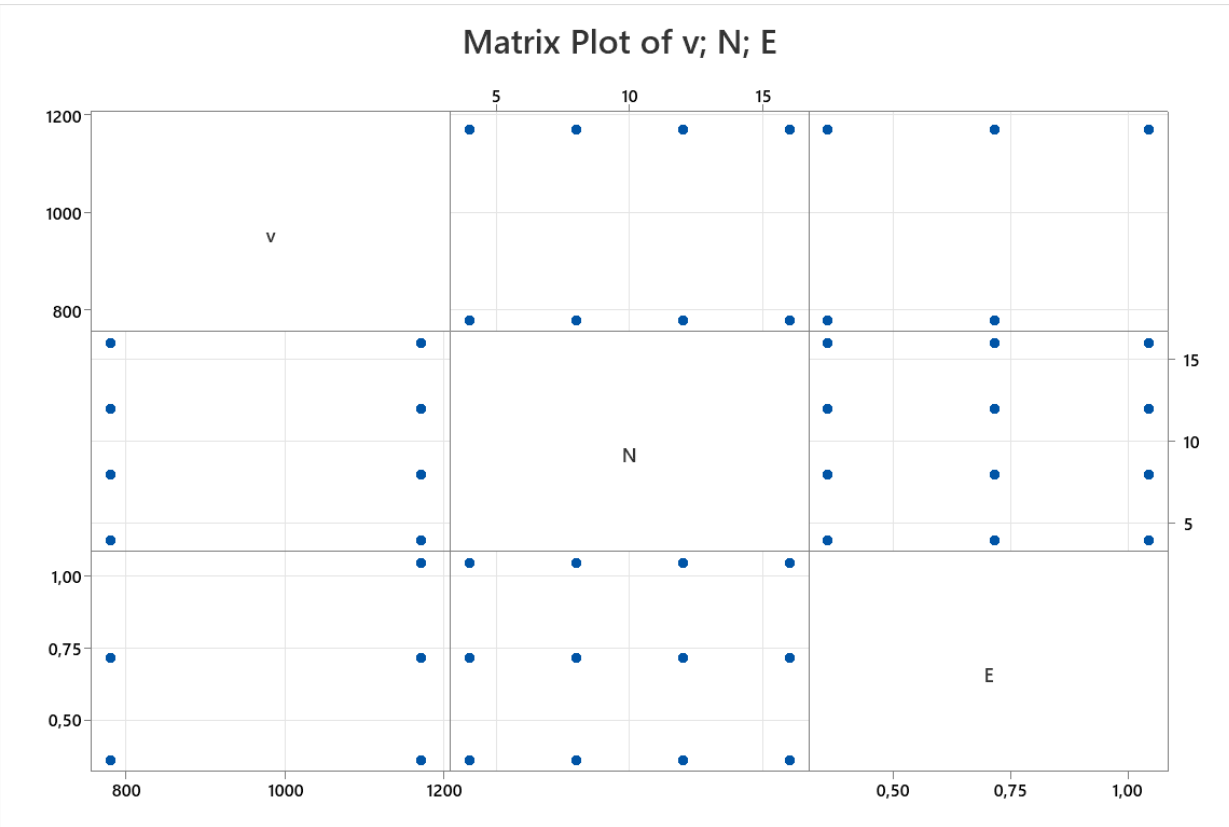
Exercise 2

The laser engraving process is under study. The objective of the work is to evaluate the influence of process parameters, i.e scanning speed v (mm/s), pulse energy E (mJ) and number of passes N , on the depth of removed material, d (μm). The data are saved in the “Engraving.csv” file.

1. Suggest a model for the data. Use a stepwise regression approach.

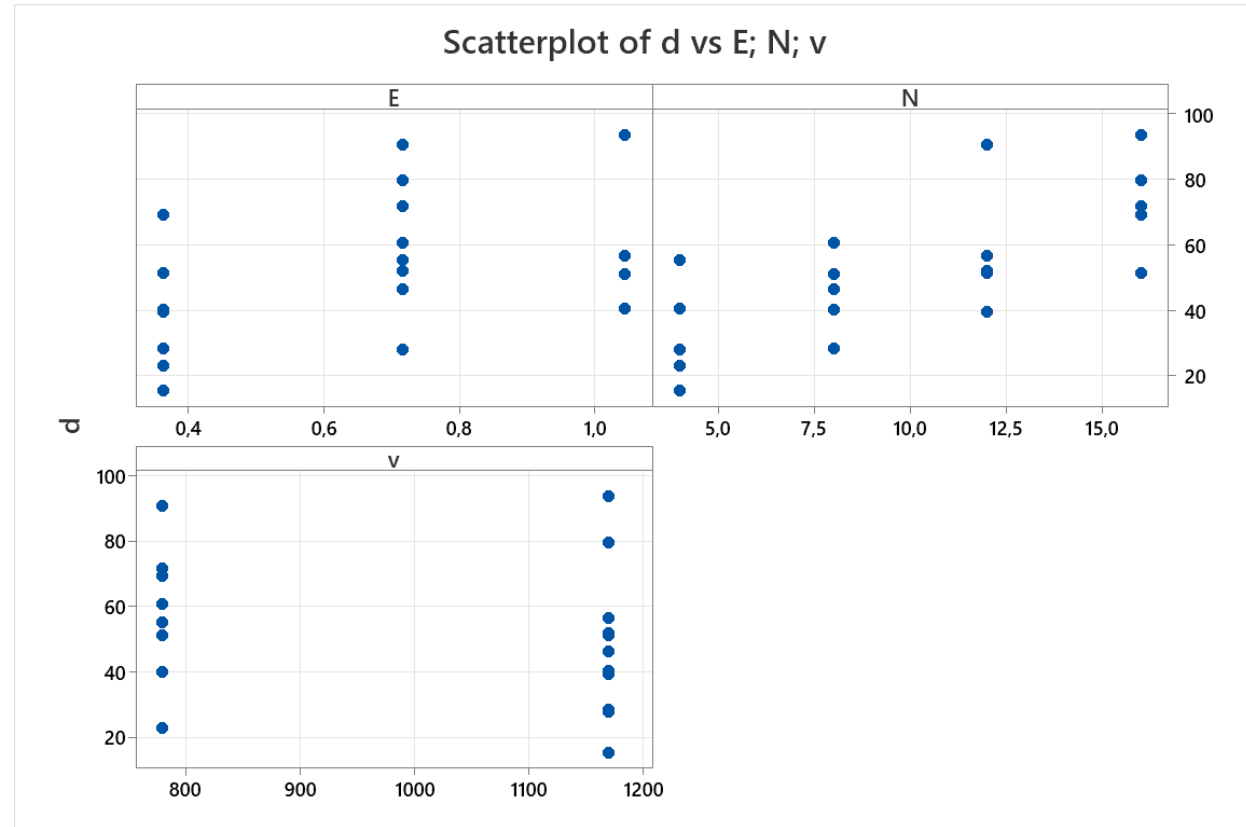
2.1 Suggest a model for the data.

Graph $\rightarrow$ Matrix Plot $\rightarrow v, N, E$



The regressor are placed in a specific order, indicating a planned design of the experiment.

*Graph $\rightarrow$ Scatterplot (d vs v , d vs N , d vs E)
Multiple graphs: Separate panels*



A linear relationship between d and N is visible. No clear pattern for d vs E and d vs v .

2.1 Suggest a model for the data.

Stat → Regression → Fit regression model

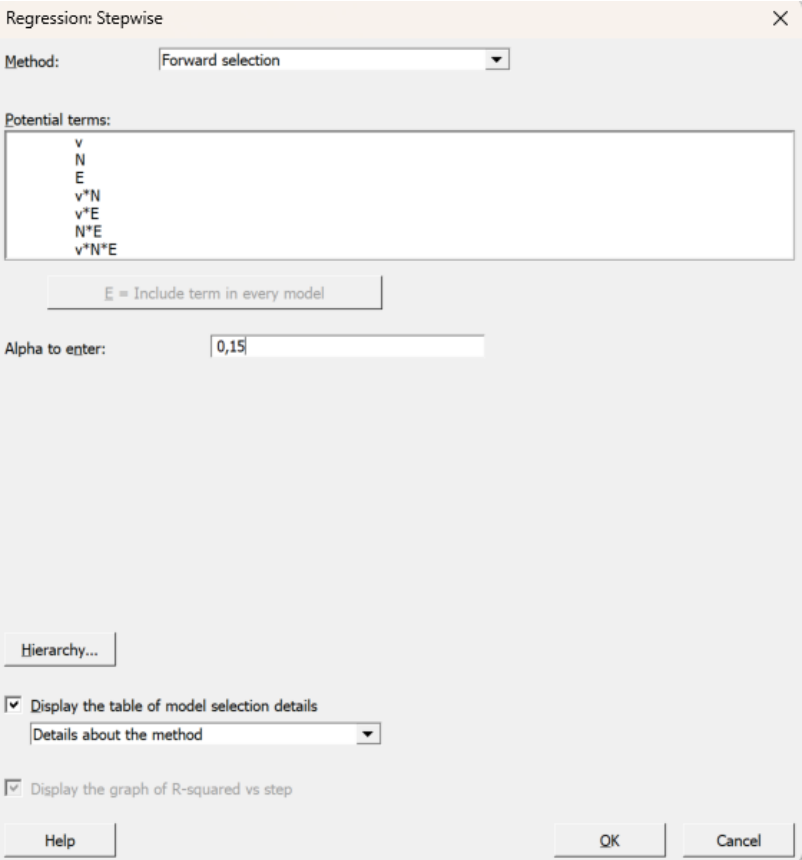
Response = d

Continuous predictors= v, N, E

→ Model = interactions up to third-order

→ Stepwise = Forward Selection, alpha to enter = 0,15

→ Storage = Fits, Residuals



Regression Equation

$$d = 33,1 - 0,0431 v + 3,372 N + 45,26 E$$

Analysis of Variance

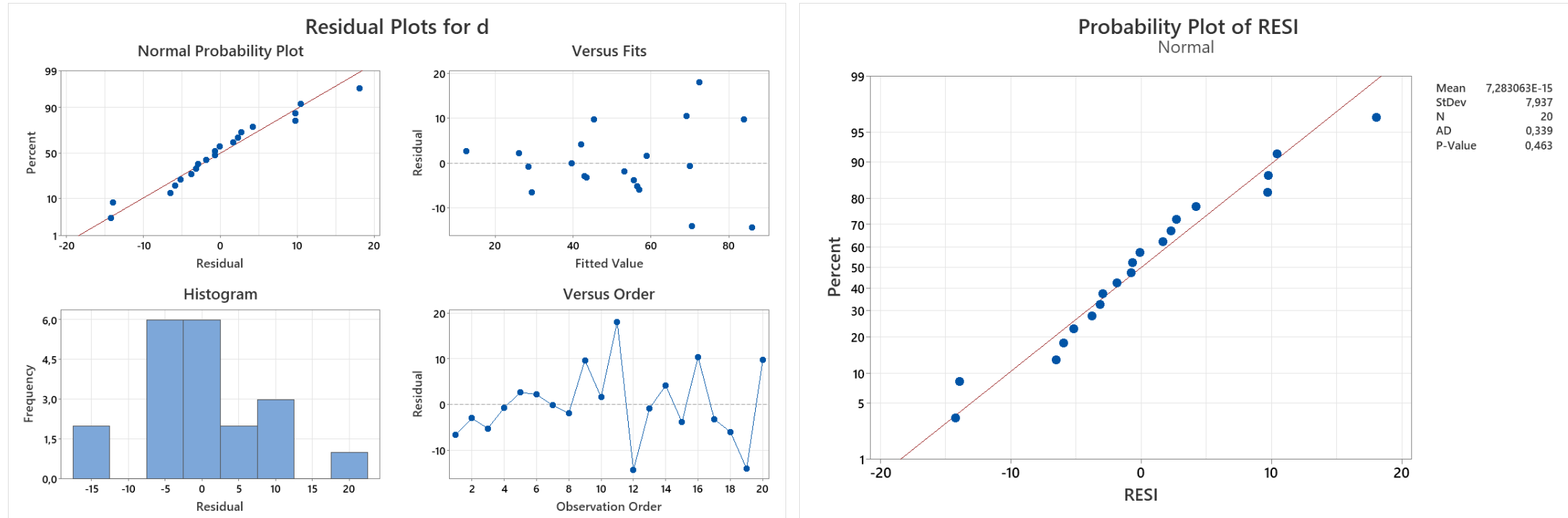
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	7370	2456,79	32,84	0,000
v	1	1217	1217,03	16,27	0,001
N	1	4549	4548,96	60,80	0,000
E	1	2414	2414,25	32,27	0,000
Error	16	1197	74,82		
Total	19	8567			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
8,64968	86,03%	83,41%	78,01%

2.1 Suggest a model for the data.

Residual analysis



The normality assumption of the residuals cannot be rejected. An additive model is the selected model for the data.